

Darrel Feiro Wihandi

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Education

University of Waterloo

BSE Software Engineering Candidate

Relevant Coursework: Data Structures & Data Management, Digital Computers, Database Management, Software Engineering Principles, Combinatorics, Compilers

Skills

Languages: C++, Python, Rust, HLSL, WGSL, TypeScript

Graphics & Compilers: SPIR-V, Vulkan, WebGPU, LLVM, CUDA, RenderDoc, CMake

Systems & Backend: Linux, Git, gRPC, PostgreSQL, Redis, AWS, Docker, Kubernetes

Experience

Software Engineering Intern | C++, Python, gRPC, PostgreSQL, Redis

Sept. 2026 – Present

Carta

- Built a **self-service enterprise SSO workflow** with C++ gRPC APIs for configuration CRUD operations, replacing staff-assisted setup and reducing estimated onboarding time from **1–2 hours to within 15 minutes**.
- Extended enterprise SSO integration with **SCIM provisioning** for user lifecycle and group membership changes, propagating identity updates across downstream services.
- Built a **Python MockIdP** to validate **SSO and SCIM provisioning workflows**, provisioning test identity providers and simulating authentication responses for automated and manual testing of **SSO enforcement and portfolio transfers**.

Graphics Compiler Engineer | C++, Slang, Shader Compilation

Feb. 2025 – Present

NVIDIA Slang – Open Source Contributor

- Extended Slang's **SPIR-V and WGSL code generation for texture-dimension queries**, supporting 2D textures, array layers, and mip-level operations.
- Debugged **cross-backend compilation inconsistencies** involving mip-level arguments and integer result types, tracing failures through generated shader code.
- Developed **regression tests validating texture-query behavior** across shader targets, including array textures and multiple mip levels.

Machine Learning Systems Engineer | Python, OpenTelemetry, SQLite

May 2026 – Oct. 2026

Wat.AI – TRACE

- Built **TRACE**, an AI-agent reliability framework that pinpoints failures in multi-step workflows and generates reproducible test cases, reducing investigation time by **40%**.
- Engineered **Python trace analysis and deterministic state validation** to distinguish root causes from downstream errors and produce evidence-backed failure reports.
- Automated **500+ evaluation runs** across **30+ adversarial scenarios**, integrating regression replay into CI to catch recurring failures and verify fixes preserve valid agent behavior.

AI Software Engineer | Python, Azure OpenAI, FastAPI

Jan. 2026 – May 2026

Rogers Communications

- Built a **Python pipeline for spectrum acquisition analysis**, structuring regulatory filings and combining them with regional customer data to help the strategy team prioritize frequency bands and markets for acquisition review.
- Developed a **metadata enrichment application** integrating Colibra APIs, Azure OpenAI, and business-glossary retrieval, with validation and human review; reduced authoring time from **6 to 2 minutes per asset**.

Software Engineer | Python, AWS, Kubernetes

May 2025 – Aug. 2025

Eon Media

- Deployed **GPU-accelerated video processing services on AWS EKS**, configuring GPU resources, health checks, and S3 storage; automated CI/CD across EKS and EC2, reducing deployment overhead by **50%**.
- Implemented **C++/CUDA kernels for fused frame resizing and color conversion** in a real-time sports analytics pipeline, benchmarking preprocessing latency against OpenCV ahead of YOLO inference.

Projects

Flux ⚡ | C++, SPIR-V, WebGPU, WebAssembly

Feb. 2026 – Present

Shader Language, Compiler, and Interactive GPU Playground

- Designed a **typed shader language and compiler**, translating source programs through a custom intermediate representation and optimization passes into SPIR-V and WGSL.
- Built an **interactive WebGPU playground** for editing and previewing shaders, with live diagnostics and a WebAssembly visualizer showing how source changes affect compiled output.